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Task Description:

Our research intends to study multiphase flow behavior related to human lungs in microgravity by both theoretical and experimental means. Since the lung's airways are internally coated with a thin liquid film, closure of the airway can occur by a capillary instability resulting in formation of a liquid plug. This is most often found near the end of expiration, but sooner under various abnormal conditions. A closed airway cannot participate in gas exchange. In normal gravity, closure is limited to the lower lung due to the body force on the tissues, while the upper lung remains open. In microgravity, however, it can be more uniformly distributed. Once closed we are interested in how to reopen the airway. This process involves propagation of the formed liquid plug distally during an inspiration. As the plug deposits its contents on the airway walls, it can eventually rupture and gas exchange is again possible. Our project examines the dynamics of the capillary instability responsible for airway closure and how potentially one could prevent closure from occurring. The system is mathematically modeled and solved using a combination of asymptotic and numerical methods. The role of fluid properties, wall elasticity, surface charges, imposed oscillations and surfactants in potential stabilization of the closure process will be assessed. In addition, the process of airway reopening due to the dynamics of plug propagation will be examined through a mathematical model of two-phase flow in a tube with a similar parameter variation as the closure problem. A sequence of experiments is proposed to evaluate the effects of modifying the system parameters. The closure and subsequent reopening of lung airways due to capillary-elastic instabilities contributes strongly to the mechanical response of the lung during breathing. Understanding the underlying fluid mechanical phenomena is a fundamental inquiry in lung physiology and function, both in microgravity situations and on earth.
The liquid plug propagation which occurs in airway reopening is also a phenomenon seen in other pulmonary applications. For example, treatment of certain diseases involves instillation of drugs mixed with liquid directly into the trachea and forced throughout the lung by imposed ventilation. This is the common mode of delivering surfactants into the lungs of prematurely-born infants who suffer from surfactant deficiency (hyaline membrane disease, also called respiratory distress syndrome of the newborn). Liquid plugs are formed during these procedures and blown distally by mechanical inspiration. The resulting spatial distribution in the lung depends on the system parameters governing the basic plug flow phenomenon that we study. Delivery of genetic material and other drugs into the lung can be done by liquid instillation techniques, too.

Another pulmonary example is liquid ventilation, a means of ventilating a critically ill patient with perfluorocarbon liquid. The process of filling the lung with the perfluorocarbon, as well as the development and motion of plugs are essential to the effectiveness of the treatment and depend on the liquid plug flows we are examining. In addition to the pulmonary application, this research addresses issues which are basic to other engineering fields. In microfluidic devices which involve gas and liquid flows, it is common to have plug propagation as an underlying means of moving samples through the chip. The length scales for these devices is quite similar to the small airways of the lung, so our results are directly applicable to this rapidly growing field. In manufacturing there are two air-liquid free surfaces in annular extrusion methods for creating small hollow fibers. These may be materials processed for a number of applications. Also, there are methods of co-extrusion manufacturing (e.g. optical fibers) in which there is an inner plug core of liquid and an outer annulus of a different liquid extruded simultaneously, one around the other. Also, these extrusions do not remain entirely fluid. Even though initially both surfaces in these applications are fluid-fluid interfaces, the outer surface in extruded forms will cool before the inner one, and that surface will develop flexible, elastic properties during part of the cooling process.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

In Halpern & Grotberg, 2003 (1) we investigate the stability of core-annular film flows (CAFFs) consisting of two immiscible fluids in a cylindrical tube with circular cross-section. CAFFs are important in many industrial and biomedical applications. In secondary oil recovery, for example, water is pumped into the well to displace the remaining oil. It is also significant in the lung, where a thin liquid film coats the inner surface of the small airways of the lungs. CAFFs are influenced by a surface-tension instability, which may induce the breakup of the core fluid into short plugs, reducing the efficiency of oil recovery, or blocking the passage of air in a lung. We consider the stability of a thin film coating the inner surfaceof a rigid tube with a circular cross-section and with the less viscous fluid occupying the core. In the absence of a core flow, for thick enough films, the surface tension instability promotes a liquid bulge that can grow to eventually create a plug blocking the tube. The current analysis explores the effects of an oscillatory core flow in the high frequency limit which enables us to uncouple the core and film problems. Lubrication theory is used to derive a nonlinear evolution equation for the position of the air-liquid interface. Numerical solutions demonstrate that the core flow can prevent plug formation of the more viscous film layer by nonlinear saturation of the surface tension instability. The stabilization mechanism is akin to that of a reversing butter knife, where the core shear wipes the growing liquid bulge back on to the tube wall during the main tidal volume stroke, but allows it to grow back as the stroke and shear turn around.
In Anderson, Molthen , Dawson , Haworth , Bull , Glucksberg, and Grotberg, 2004 (2) we studied the process of liquid instillation into the pulmonary airways during medical procedures such as surfactant replacement therapy, partial liquid ventilation, and pulmonary drug delivery. For all cases, understanding the dynamics of liquid distribution in the lung will increase the efficacy of treatment. A recently developed imaging technique for the study of real-time liquid transport dynamics in the pulmonary airways was used to investigate the effect of respiratory rate on the distribution of an instilled liquid surfactant, in a rat lung. Twelve excised rat lungs were suspended vertically, and a single bolus (0.05 ml) of exogenous surfactant (Survanta, Ross Laboratories, Columbus, OH) mixed with radiopaque tracer was instilled as a plug into the trachea. The lungs were ventilated with a 4-ml tidal volume for 20 breaths at one of two respiratory rates: 20 or 60 breaths/min. The motion of radiodense surfactant was imaged at 30 frames/s with a microfocal X-ray source and an image intensifier. Dynamics of surfactant distribution were quantified for each image by use of distribution statistics and a homogeneity index. We found that the liquid distribution depended on the time to liquid plug rupture, which depends on ventilation rate. At 20 breaths/min, liquid was localized in the gravity-dependent region of the lung.At 60 breaths/min, the liquid coated the airways, providing a more vertically uniform liquid distribution.

In Halpern, Wei, and Grotberg, 2003 (3) we consider the effects of time-periodic pressure-driven flows on the stability of a coreannular flow with insoluble surfactant present at the fluid-fluid interface. In the limit of a thin film coating the inner surface of a rigid tube with a circular cross section, we obtained a pair of coupled evolution equations with time-periodic coefficients for the perturbed fluid-fluid interface and for the interfacial surfactant concentration. For a steady base flow, the interaction between surfactant and the base flow may cause the interfacial instability to grow at a faster rate than the stationary base state case. Time periodic base flows can alter the features of the steady-based

Task Progress:	instability depending on the magnitude of the surface tension, surfactant activity, andoscillatory frequency. For an oscillatory base flow with zero mean, the growth rate decreases monotonically as the frequency increases. At low frequencies, the growth rate reaches a maximum corresponding to the growth rate for a steady base flow having the same amplitude. At sufficiently high frequencies, the growth rate reaches a minimum corresponding to the growth rate in the limit of a stationary base state. The linear stability can be readily extended to other time-periodic base flows. In Fujioka and Grotberg, 2004 (4, 5) we study the steady propagation of a liquid plug within a two-dimensional channel lined by a uniform liquid film. A finite volume scheme is used to solve the Navier-Stokes equations numerically. We examine the effect of plug propagation speed and plug length. For a fixed plug length, the trailing film thickness increases with plug propagation speed. Our model agrees with previous theories for semi-infinite bubble propagation if the plug length is greater than the channel width. The trailing film thickness decreases as the plug length decreases below the channel width, and for finite Re there is a significant interaction between the leading and trailing menisci. A recirculation flow forms inside the core of the plug, which, as Re increases, is tilted towards the rear meniscus. The recirculation velocity between both tips decreases with the plug length. The macroscopic pressure gradient, defined as the pressure drop between the leading and trailing gas phases divided by the plug length, is a function of U and U2, where U is the plug propagation speed, when the fluid properties and channel geometry are unchanged. The U2 term becomes dominant at small values of the plug length. A capillary wave develops at the front meniscus, with an amplitude that increases with Re, and can cause large local changes in wall shear stresses and pressures. Refereed Journal Publications (Oct. 1, 2003 through Sept. 30, 2004). 1. Halpern, D. and Grotberg, J.B. Nonlinear Saturation of the Rayleigh instability in a liquid-lined tube due to oscillatory flow. J. Fluid Mech. 492: 251-270, 2003. 2. Anderson JC, Molthen RC, Dawson CA, Haworth ST, Bull JL, Glucksberg, MR and JB Grotberg. Effect of ventilation rate on instilled surfactant distribution in the pulmonary airways of rats. J Appl Physiol 97: 45-56, 2004. Abstracts and Presentations (Oct. 1, 2003 through Sept. 30, 2004). 3. Halpern, D., Wei, H.H. and Grotberg, J.B. The Linear Stability of a Time-Periodic Core-Annular Flow in the Presence of Surfactant. American Physical Society, Division of Fluid Dynamics Annual Meeting, Meadowlands, New Jersey, November 23-25, 2003. 4. Fujioka, H. and Grotberg, J.B. The propagation of a surfactant laden liquid plug in a 2D-channel, Biomedical Engineering Society Annual Meeting, Nashville, October 1-4, 2003. 5. Fujioka, H. and Grotberg, J.B. The propagation of a surfactant laden liquid plug in a 2D-channel, American Physical Society, Division of Fluid Dynamics Annual Meeting, Meadowlands, New Jersey, November 23-25, 2003
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Articles in Peer-reviewed Journals	Anderson JC, Molthen RC, Dawson CA, Haworth ST, Bull JL, Glucksberg MR, Grotberg JB. "Effect of ventilation rate on instilled surfactant distribution in the pulmonary airways of rats." J Appl Physiol. 2004 Jul;97(1):45-56. Epub 2004 Feb 27. PMID: 14990558 , Jul-2004